SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO5: Apply association rule mining techniques to discover interesting relationships and patterns within large datasets
Assessment Tool Weightage: 35%

QUESTIONS: 05 **Total Marks:50**

Annexure A

Pseudocode Writing Task (Algorithm Design)

Criteria	Problem Understanding *(2)*	Algorithm Design *(3)*	Use of Control Structures *(2)*	Pseudocode Structure *(1)*	Completeness *(2)*	***Total(10)***
Weightage	*20%*	30%	20%	10%	20%	*100%*
Q1						
Q2						
Q3						
Q4						
Q5						

Code of Conduct:

I G Madhulatha(192311295) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
G Madhulatha

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20%	Identifies all inputs, outputs, constraints accurately	Minor missing elements	Partial understanding	Incorrect interpretation
Algorithm Design	30%	Complete, correct, and logically sound solution	Minor logical gaps	Partially correct logic	Incorrect approach
Use of Control Structures	20%	Efficient and appropriate use of loops, conditions, functions/modules	Minor inefficiencies	Limited or basic usage	Incorrect or missing constructs
Pseudocode Structure	10%	Well-structured, clear, consistent format, easy to follow	Mostly clear with minor issues	Difficult to follow in parts	Poorly structured and unclear
Completeness	20%	Handles all cases; optimized approach	Handles most cases; reasonable efficiency	Handles basic cases only	Incomplete or inefficient

Annexure A

Q1. Dataset: Online Shopping Transactions

Order ID	Products Purchased
O1	Mobile, Earphones
O2	Mobile, Charger
O3	Earphones, Charger
O4	Mobile, Earphones, Charger
O5	Mobile, Earphones

Question:
Design pseudocode for the Apriori algorithm to discover frequent product combinations from the online shopping dataset using a specified minimum support threshold. Clearly illustrate the candidate itemset generation and pruning process at each iteration. (BL3 – 1.05 Marks)

Q2. Dataset: Library Book Borrowing Records

Borrow ID	Books Borrowed
B1	Data Science, Python
B2	Python, Machine Learning
B3	Data Science, Machine Learning
B4	Data Science, Python, Machine Learning
B5	Python, Machine Learning

Question:
Develop pseudocode for the FP-Growth algorithm to identify frequent borrowing patterns without generating candidate itemsets. Explain the steps involved in constructing the FP-Tree using the given borrowing records. (BL4 – 1.05 Marks)

Q3. Dataset: Frequently Purchased Services

Service Combination	Support Count
{Internet}	4
{Streaming}	4
{Cloud Storage}	3
{Internet, Streaming}	3
{Streaming, Cloud Storage}	2

Question:
Write pseudocode to generate association rules from the given frequent service combinations. Include the procedures for confidence calculation, rule generation, and filtering based on a minimum confidence threshold. (BL3 – 1.05 Marks)

Q4. Dataset: Student Course Enrollment

Student ID	Courses Enrolled
S1	Python, Database
S2	Python, AI
S3	Database, AI
S4	Python, Database, AI
S5	Python

Question:
Design pseudocode for constraint-based frequent itemset mining where only itemsets containing the course "Python"are considered during the mining process. Explain how the constraint affects candidate generation and pruning. (BL4 – 1.05 Marks)

Q5. Dataset: Hospital Treatment Hierarchy

Patient ID	Treatments Received
P1	Healthcare, Cardiology, ECG
P2	Healthcare, Orthopedics
P3	Healthcare, Cardiology

P4	Healthcare, Cardiology, ECG
P5	Healthcare, Orthopedics

Question:
Develop pseudocode for multilevel association rule mining using the treatment hierarchy provided in the dataset. Explain how rules are generated at different hierarchy levels and how concept hierarchies improve pattern discovery.

ASSOCIATION RULE MINING – PSEUDOCODE

Q1. Dataset: Online Shopping Transactions

Given Data

O1 – Mobile, Earphones
O2 – Mobile, Charger
O3 – Earphones, Charger
O4 – Mobile, Earphones, Charger
O5 – Mobile, Earphones

Question

Design pseudocode for the Apriori algorithm to discover frequent product combinations using a specified minimum support threshold. Clearly illustrate candidate itemset generation and pruning.

Answer

Apriori is a level-wise algorithm that generates candidate itemsets and removes candidates that do not satisfy the minimum support threshold.

Pseudocode

```
BEGIN
Input:
Transaction database D
Minimum support threshold minsup
Step 1:
Scan D and find all individual products.
C1 = all 1-item candidate sets.
Step 2:
Count the support of every item in C1.
Step 3:
Remove items whose support is less than minsup.
L1 = frequent 1-itemsets.
Step 4:
Set k = 2.
Step 5:
Repeat while L(k-1) is not empty:
  Generate candidate itemsets Ck from L(k-1).

  For every pair of itemsets in L(k-1):
    Join them if their first k-2 items are identical.

  For every candidate c in Ck:
    If any (k-1)-subset of c is not present in L(k-1):
      Remove c from Ck.

  Scan the transaction database D.

  Count the support of every remaining candidate in Ck.

  Remove candidates whose support is less than minsup.

  Lk = remaining frequent k-itemsets.

  k = k + 1.
Step 6:
Combine all frequent itemsets:
L = L1 ∪ L2 ∪ L3 ∪ ...
Output:
All frequent product combinations.
END
```

Candidate Generation and Pruning Example

The products are:
Mobile, Earphones, Charger.
Therefore, the initial candidate set is:
C1 = {Mobile}, {Earphones}, {Charger}
After calculating support, itemsets below the minimum support are removed to obtain L1.
Next, pairs are generated:
C2 = {Mobile, Earphones}
{Mobile, Charger}
{Earphones, Charger}

Again, support is calculated and itemsets below the minimum support are pruned.
If all required 2-itemsets are frequent, 3-itemset candidates can be generated:
C3 = {Mobile, Earphones, Charger}
The candidate is retained only if its support satisfies the minimum support threshold.
Apriori Pruning Principle
If an itemset is infrequent, all of its supersets are also infrequent.
Therefore, Apriori avoids generating unnecessary candidates and reduces the search space.

Q2. Dataset: Library Book Borrowing Records
Given Data
B1 – Data Science, Python
B2 – Python, Machine Learning
B3 – Data Science, Machine Learning
B4 – Data Science, Python, Machine Learning
B5 – Python, Machine Learning
Question
Develop pseudocode for the FP-Growth algorithm to identify frequent borrowing patterns without generating candidate itemsets. Explain the steps involved in constructing the FP-Tree using the given borrowing records.
Answer
FP-Growth is different from Apriori because it does not generate candidate itemsets. It stores the transactions in a compressed FP-Tree and mines frequent patterns directly from the tree.
Pseudocode

```
BEGIN
Input:
Transaction database D
Minimum support threshold minsup
Step 1:
Scan D once.
Count the frequency of every item.
Step 2:
Remove items whose support is less than minsup.
Step 3:
Sort the remaining items in descending order of frequency.
Step 4:
Create an empty FP-Tree with a root node.
Step 5:
For each transaction T in D:
    Remove infrequent items.

    Sort the remaining items according to
    the global frequency order.

    Insert the ordered items into the FP-Tree.

    Increase the count of existing nodes.

    Create new nodes when necessary.

    Connect nodes containing the same item
    using node links.
Step 6:
Create the header table containing:
    Item
    Frequency
    Node link
Step 7:
Start mining from the least frequent item.
For each item:
    Find its conditional pattern base.

    Construct its conditional FP-Tree.

    Recursively mine the conditional FP-Tree.
Step 8:
Combine the discovered patterns.
Output:
All frequent itemsets.
END
```

FP-Tree Construction for the Given Dataset
First count the items.

Data Science occurs in:
B1, B3, B4
Frequency = 3
Python occurs in:
B1, B2, B4, B5
Frequency = 4
Machine Learning occurs in:
B2, B3, B4, B5
Frequency = 4
Therefore, the frequency order can be:
Python → Machine Learning → Data Science
Transactions are reordered according to this frequency.
B1:
Python → Data Science
B2:
Python → Machine Learning
B3:
Machine Learning → Data Science
B4:
Python → Machine Learning → Data Science
B5:
Python → Machine Learning
These ordered transactions are inserted into the FP-Tree.
Common prefixes are shared, which compresses the database.
For example, B2 and B5 both contain:
Python → Machine Learning
Therefore, the same branch can be shared and its count is increased.
Main Advantage
FP-Growth does not repeatedly generate candidate itemsets. It uses the compressed FP-Tree and conditional FP-Trees to discover frequent patterns efficiently.

Q3. Dataset: Frequently Purchased Services
Given Data
{Internet} – Support Count = 4
{Streaming} – Support Count = 4
{Cloud Storage} – Support Count = 3
{Internet, Streaming} – Support Count = 3
{Streaming, Cloud Storage} – Support Count = 2
Question
Write pseudocode to generate association rules from the given frequent service combinations. Include the procedures for confidence calculation, rule generation, and filtering based on a minimum confidence threshold.
Answer
Association rules are generated from frequent itemsets.
For example:
{Internet, Streaming}
can generate:
Internet → Streaming
Streaming → Internet
Confidence Formula
For a rule:
X → Y
Confidence is calculated as:
Confidence(X → Y) =
Support(X ∪ Y) / Support(X) × 100
Pseudocode
```
BEGIN
Input:
Frequent itemsets F
Minimum confidence threshold minconf
For each frequent itemset I in F:
If the size of I is at least 2:

    Generate all non-empty proper subsets X of I.

    For each subset X:

        Y = I - X

        Calculate:
```

confidence =
support(I) / support(X) × 100

If confidence >= minconf:

Generate rule:

X → Y

Add the rule to the result.

Output:
All association rules satisfying minconf.
END

Example 1

Frequent itemset:
{Internet, Streaming}
Support count = 3
Internet support count = 4
Therefore:
Confidence(Internet → Streaming)
= 3/4 × 100
= 75%
Thus, if minimum confidence is 70%:
Internet → Streaming
is a valid association rule.

Example 2

Confidence(Streaming → Internet)
= 3/4 × 100
= 75%
Therefore:
Streaming → Internet
is also valid when minimum confidence is 70%.

Example 3

Frequent itemset:
{Streaming, Cloud Storage}
Support count = 2
Streaming support count = 4
Confidence(Streaming → Cloud Storage)
= 2/4 × 100
= 50%
If minimum confidence is 70%, this rule is rejected.
Cloud Storage support count = 3.
Confidence(Cloud Storage → Streaming)
= 2/3 × 100
= 66.67%
This is also rejected when minimum confidence is 70%.

Final Point

The minimum confidence threshold filters out weak rules and retains only strong relationships between frequently purchased services.

Q4. Dataset: Student Course Enrollment

Given Data

S1 – Python, Database
S2 – Python, AI
S3 – Database, AI
S4 – Python, Database, AI
S5 – Python

Constraint

Only itemsets containing the course "Python" should be considered.

Question

Design pseudocode for constraint-based frequent itemset mining where only itemsets containing "Python" are generated during the mining process. Explain how the constraint affects candidate generation and pruning.

Answer

The constraint is:
Python must be present in every itemset.
Therefore, itemsets that do not contain Python are not generated or considered.

Pseudocode

BEGIN

Input:
Transaction database D
Minimum support threshold minsup
Constraint:
Every itemset must contain "Python".
Step 1:
Find the support of Python.
Step 2:
If Python does not satisfy minsup:
    Stop.

    No valid itemset can be generated.
Step 3:
Generate candidate itemsets containing Python.
C1 = {Python}
Step 4:
Count support of Python.
If support(Python) >= minsup:

    Add {Python} to frequent itemsets.
Step 5:
Generate larger candidates only by combining
Python with other items.
Example:

{Python, Database}
{Python, AI}
Step 6:
Calculate support for each candidate.
Step 7:
Prune any candidate that:
    a) Does not contain Python, or
    b) Does not satisfy minimum support.
Step 8:
Generate larger itemsets only if they contain Python.
Example:

{Python, Database, AI}
Step 9:
Continue until no more frequent candidates can be generated.
Output:
Frequent itemsets containing Python.
END

Applying the Constraint to the Dataset

Python occurs in:
S1, S2, S4 and S5.
Therefore:
Support(Python) = 4/5 × 100 = 80%
The possible Python-containing 2-itemsets are:
{Python, Database}
Occurs in S1 and S4.
Support = 2/5 × 100 = 40%
{Python, AI}
Occurs in S2 and S4.
Support = 2/5 × 100 = 40%
The Python-containing 3-itemset is:
{Python, Database, AI}
Occurs only in S4.
Support = 1/5 × 100 = 20%

Effect of the Constraint

Without the constraint, the algorithm may generate:
{Database}, {AI}, {Database, AI}, etc.
With the constraint, these itemsets are immediately excluded because they do not contain Python.
Therefore, candidate generation and pruning become more efficient.

Final Point

The constraint reduces the search space by forcing every candidate to contain Python. This avoids generating irrelevant itemsets and makes frequent itemset mining faster.

Q5. Dataset: Hospital Treatment Hierarchy

Given Data

P1 – Healthcare, Cardiology, ECG
P2 – Healthcare, Orthopedics
P3 – Healthcare, Cardiology
P4 – Healthcare, Cardiology, ECG
P5 – Healthcare, Cardiology

Treatment Hierarchy

Healthcare
↓
Cardiology / Orthopedics
↓
ECG

Question

Develop pseudocode for multilevel association rule mining using the treatment hierarchy provided in the dataset. Explain how rules are generated at different hierarchy levels and how concept hierarchies improve pattern discovery.

Answer

Multilevel association rule mining discovers associations at different levels of a concept hierarchy.
For this dataset:
Level 1:
Healthcare
Level 2:
Cardiology
Orthopedics
Level 3:
ECG

Pseudocode

BEGIN
Input:
Transaction database D
Treatment hierarchy H
Minimum support at each level minsup
Step 1:
Start at the highest level of the hierarchy.
Step 2:
Generate frequent itemsets using Level 1 concepts.
Example:

{Healthcare}
Step 3:
Calculate support for Level 1 itemsets.
Step 4:
Generate Level 2 candidates from frequent
higher-level concepts.
Examples:

{Healthcare, Cardiology}
{Healthcare, Orthopedics}
Step 5:
Calculate support of Level 2 itemsets.
Step 6:
Prune itemsets that do not satisfy minimum support.
Step 7:
Drill down to Level 3.
Generate more specific candidates.

Example:

{Healthcare, Cardiology, ECG}
Step 8:
Calculate support at Level 3.
Step 9:
Generate association rules from the frequent
itemsets at each level.
Step 10:
Continue until the lowest required hierarchy
level is reached.
Output:
Multilevel frequent itemsets and association rules.
END

Level 1 Analysis

Healthcare occurs in all five patients.

Support:
5/5 × 100 = 100%
Therefore:
{Healthcare}
is highly frequent.
Level 2 Analysis
Cardiology occurs in:
P1, P3, P4 and P5.
Support:
4/5 × 100 = 80%
Orthopedics occurs in:
P2.
Support:
1/5 × 100 = 20%
Therefore, Cardiology is more frequent than Orthopedics.
Level 3 Analysis
ECG occurs in:
P1 and P4.
Support:
2/5 × 100 = 40%
A more specific pattern can therefore be identified:
Cardiology → ECG
Both Cardiology and ECG occur together in P1 and P4.
Support:
2/5 × 100 = 40%
Example of Multilevel Rules
At a general level:
Healthcare → Cardiology
At a more specific level:
Cardiology → ECG
Thus, the algorithm can discover both broad and specific treatment relationships.
How Concept Hierarchies Improve Pattern Discovery
Concept hierarchies allow the algorithm to analyze data at different levels of detail.
At a high level, it can discover general patterns such as:
Healthcare → Cardiology
At a lower level, it can discover more specific patterns such as:
Cardiology → ECG
This provides both general and detailed knowledge and helps organizations understand patterns that may not be visible at only one level of abstraction.
Final Point
Multilevel association rule mining uses the treatment hierarchy to progressively move from general concepts to specific concepts. This improves pattern discovery by providing meaningful relationships at different levels of detail.